MIL-S-43489D
13 October 1987
SUPERSEDING
MIL-S-43489C
31 July 1974

MILITARY SPECIFICATION

SLING, BAG AND CASE CARRYING: ST-35

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of cotton webbing sling with metal hardware components.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

A-A-203	- Paper, Kraft, Untreated
V-T-285	- Thread, Polyester
DDD-L-20	- Label: For Clothing, Equipage, and Tentage (General Use)
PPP-B-636	- Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8465

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MIL-W-530	- Webbing, Textile, Cotton, General Purpose, Natural or in Colors
MIL-B-543	- Buckles, Tongueless and Web Strap
MIL-R-2327	- Rings, Connecting, Round
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-T-43566	- Tape, Textile, Cotton or Polyester, General Purpose, Natural or in Colors
MIL-S-43770/9	- Snap Hooks, Cast Body, Fixed Loop Eye, Trigger Closure
MIL-W-44049	- Webbing, Textile and Tape, Textile, Polypropylene, General Purpose, Natural or in Colors

STANDARDS

FEDERAL

FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-731	- Quality of Wood Members for Containers and Pallets

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-3-204	- Sling, Bag and Case, Carrying. ST-35; Assembly

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Samples.

3.1.1 Guide samples. Samples of the end item, when furnished, are solely for guidance and information to the manufacturer (see 6.3). Variation from this document may appear in the sample, in which case the document shall govern.

* 3.1.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.2 Materials (see 6.5).

3.2.1 Webbing.

* 3.2.1.1 Webbing, 2-inch. The 2-inch webbing shall be cotton conforming to type II, class 4 of MIL-W-530, or as an alternate, the webbing shall be polypropylene conforming to type II, class 4 of MIL-W-44049.

* 3.2.1.2 Webbing, 1-inch. The 1-inch webbing shall be cotton conforming to type III, class 4 of MIL-W-530, or as an alternate, the webbing shall be polypropylene conforming to type III, class 4 of MIL-W-44049.

3.2.1.3 Webbing, 3/4-inch. The 3/4-inch webbing shall be cotton conforming to type IIa, class 4 of MIL-W-530, or as an alternate, the webbing shall be polypropylene conforming to type IIa, class 4 of MIL-W-44049.

3.2.1.4 Color. The color of the webbing shall be Olive Drab 7.

3.2.2 Tape, cotton. The cotton tape shall be type I, class 4, 1 inch conforming to MIL-T-43566. The color of the tape shall be Olive Drab 7.

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3.2.3 Thread, polyester. The polyester thread shall be type I, class 1, of sub-class A, size FF conforming to V-T-285. Thread shall be dyed OD S-1 C.A. 66022 and shall show fastness to weathering equal to or better than the standard sample. When no standard sample is available, the thread shall show good fastness to weathering.

3.2.4 Buckle. The steel buckles shall be tongueless, one-bar, type I, style 2, class 3, construction A, sizes 3/4 inch and 1 inch conforming to MIL-B-543.

3.2.5 Snap hook. The snap hook shall be cast malleable iron, 3/4 inch conforming to MIL-S-43770/9.

3.2.6 Ring. The ring shall be steel wire, class 2, size 7 conforming to MIL-R-2327.

3.3 Construction. The construction of the sling shall conform in all respects to the drawing listed in section 2 and shall be as specified herein.

3.3.1 Stitches, seams, and stitchings. All stitches, seams, and stitchings shall be performed using stitch type 301 of FED-STD-751, with 6 to 8 stitches per inch and 1/8 inch stitching margin.

3.3.1.1 Automatic stitching. Any of the stitching operations cited in this document may be performed with an automatic machine provided the requirements for stitch pattern, stitches per inch, and size and types of thread are met; and at least three tying, overlapping or backstitches are used to secure the ends of stitching.

3.3.1.2 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched 1/2 inch minimum except when caught in other seams or stitching. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.3.1.3 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or runoff stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

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1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.3.1.4 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

3.4 Marking. The identification marking shall be applied in the location shown on the drawing and shall conform to type IV, class 5 of DDD-L-20. The marking "ST-35" shall be applied in the size characters and in the location indicated on the drawing and shall conform to type IV, class 9 of DDD-L-20. Fastness for the class 9 marking shall be as specified for class 5 marking.

3.5 Workmanship. The finished sling with assembled components as required shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels (AQL).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.3), it shall be examined for the defects specified in 4.4.3. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this document or applicable purchase document.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether cut lengths for components listed below and construction details are in accordance with specified requirements. When ever non-conformance is noted, corrections shall be made to the items affected and the process. Items which cannot be corrected shall be removed from production.

Component	Drawing
Item number 5; 3/4 inch width strap	2-3-204
Item number 3; 1 inch width strap	2-3-204

4.4.3 End item visual examination. The end item shall be examined for the defects listed below. The lot size shall be expressed in units of slings. The sample unit shall be one sling. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defects
Webbing and tape	Any hole, cut, tear, smash, or slub. Not firmly and tightly woven, edges frayed or scalloped. Multiple floats or abrasion mark. Broken or missing end or pick. Spliced, i.e., not one continuous piece.
Hardware	Broken or malformed, failing to serve intended purpose, finish partly or totally omitted, corroded area, burr, or sharp edge.

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Examine	Defect
Hardware (continued)	Not construction, type, or style specified. Not formed as specified. Spring portion of snap hook loose or defective causing spring to jam.
Open seams	Any.
NOTE:	A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or runoff stitches occur.
Seam and stitch type	Wrong seam or stitch type.
Stitching	Incomplete or any row missing.
Stitches per inch	Less than minimum specified. More than maximum specified.
Stitching margin	Not as specified.
Stitching ends	Backstitched less than 1/2 inch.
Thread breaks	Backstitched less than 1/2 inch.
NOTE:	Thread break not backstitched shall be classified as an open seam.
Components and assembly	Any component part or required operation omitted (unless otherwise classified herein). Mend or darn. Any component reversed, i.e., not assembled to body as indicated on drawing. Needle chews.
Ends	Not finished as specified. Abutted ends of binding more than 1/16 inch apart.
Marking	Omitted, incorrect, illegible, misplaced, or size characters not as specified.
Cleanness	Grease or oil stains clearly noticeable.
Thread ends	Not trimmed to 1/4 inch maximum length.

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presence of popliteal, prescapular, prefemoral, or any

* 4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Content	Number of slings per shipping container is more or less than specified. 1/

1/ One shipping container from the sample shall be examined for this defect.

4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

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plastic membrane) measuring 0.5 square inches or

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each sling shall have the straps adjusted to their shortest dimension. The sling shall be neatly folded to measure approximately 9 by 2-1/4 by 1-3/4 inches, with the hardware inside the folds. The folded sling shall be secured at each end with cotton tape or twine.

5.1.2 Commercial. The slings shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. One hundred slings, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type RSC-L, grade V2s of PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed flat ten in length, two in width, and five in depth within a shipping container. Inside dimensions of each container shall approximate 23 inches in length, 18-1/2 inches in width, and 8-3/4 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. One hundred slings, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A unit packs shall be packed flat ten in length, two in width, and five in depth within a shipping container. Inside dimensions of each shipping container shall approximate 23 inches in length, 18-1/2 inches in width, and 8-3/4 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

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5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.3 Commercial packing. Slings, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), slings, packed as specified in 5.2.2 and 5.2.3 shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 6 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable. Each tenth unit pack shall have a manila colored paper shipping tag attached for the identification information.

6. NOTES

6.1 Intended use. This sling is used in carrying various pieces of electronic type equipment.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1.2, 4.3, and 6.4).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. Type and class of unit load required (see 5.2.1).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
f. When palletization is required (see 5.3).

6.3 Samples. For access of samples, address the contracting activity issuing the invitation for bids.

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* 6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

* 6.6 Subject term (key word) listing).

Bag and Case
Electronic equipment
Sling

6.7 Changes from previous issue. The margins of this document have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-43489D	Sling, Bag and Case Carrying, ST-35

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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